OpenADMET CYP Inhibition Blind Challenge

Methodology Report

Direct inhibition (regression) and TDI (classification) · Team preheat-to-450 · Public data only

Description of the Model

Our entry is an ensemble built around two complementary views of the molecule: a 2D message-passing neural network and a 3D equivariant foundation model, both trained multitask on public CYP and ADME data. Other model families are blended in on individual isoforms where they add signal. The four isoforms differ in assay readout, data volume, and chemical coverage. In particular, CYP2D6 is measured by label-free mass spectrometry rather than fluorescence and behaves unlike the other three, so model composition and blending are set per isoform.

Models

Chemprop. A multitask directed message-passing neural network, with RDKit 2D descriptors as additional molecule-level features. Auxiliary endpoints enter as extra output heads. Ensembled across initialization seeds and model configurations.

Atom-JEPA. A publicly released SE(3)-equivariant transformer, pretrained by its authors with a joint-embedding predictive objective. We fine-tune it multitask on the same targets as Chemprop, with generated 3D conformers as input and predictions averaged over several conformers per compound. Ensembled across fine-tuning runs.

Additional model families. Two pretrained in-context learners for tabular data, TabPFN (Prior Labs) and Kumo Tabular (NVIDIA), applied to learned molecular embeddings; a tree ensemble that averages gradient boosting (LightGBM) and a random forest over molecular fingerprints and descriptors, including Mordred descriptors; and kernel ridge regression with a MinMax (count-Tanimoto) kernel on molecular fingerprints.

Training data

All training data are public, and we did no pretraining of our own. From the challenge release we used the direct-inhibition and TDI-condition dose-response data and the single-concentration primary screen, which covers far more compounds than the dose-response set and serves as an additional source of signal. We added public CYP inhibition data (PubChem qHTS and ChEMBL) and public ADME and physicochemical endpoints (LogD from AstraZeneca and ExpansionRx datasets, liver microsome clearance, efflux ratio, solubility, and plasma protein binding from Biogen dataset) as auxiliary tasks, which shape the learned representation.

Validation

Models were compared by cross-validation on the scorable compounds, using two complementary partitions: a cluster-based split that separates close analogs, and a custom split we designed to better reflect the structure of the blind test set. Blend weights are fitted out of fold. Most blind test compounds have a close analog in the training data, whereas compounds held out in cross-validation usually do not, and no split of the available data could close that gap. Cross-validation therefore asks the models to predict further from what they have seen, which

makes it a harder and somewhat less representative regime than the blind test. We used CV to compare methods, and then evaluated candidate models via the live leaderboard.

Performance Comments

Blending the 2D and 3D models consistently outperformed either one alone, both in cross-validation and on the live leaderboard. Neither dominates: the 3D model is stronger at rank ordering, while the 2D model is better on the scored metric (ST-RAE), and the two make substantially different errors. That complementarity accounts for most of the ensemble's gain, and its size varies by isoform. Once the 3D model is included, the contribution of the additional families is smaller and isoform-specific.

Multitask training on related public CYP screening data gave a clear improvement over training on the challenge data alone.

CYP2D6 is the most difficult isoform. Its errors are dominated by rank-ordering mistakes rather than by a systematic offset, and assay noise alone does not account for them.

TDI Classification Track

Time-dependent inhibition is defined by the shift in IC50 between the TDI-condition and direct-inhibition arms, so we predict it from the same multitask regression models, which output pIC50 for both arms. The predicted shift is combined with a classifier on derived features and, for CYP2D6, with information from labeled close analogs in the training data. Decision thresholds are chosen on out-of-fold predictions, because MCC is highly sensitive to the call rate at these positive rates. CYP2D6 is again the harder isoform: positives are rare, and its TDI calls inherit the ranking difficulty seen in the regression track.

No proprietary data were used in either track.